Speech to Speech Translation Systems

Benchmarking and Implementation

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April 30, 2026

Outline

Introduction

ASR: Overview

The ASR module of a S2S pipeline first converts the input speech into text. It involves several key components:

- **Audio Preprocessing:** Noise reduction and feature extraction using MFCCs, Mel spectrograms etc.
- **Encoder:** Responsible for capturing temporal dependencies and contextual information of the input audio using the extracted features. Typically implemented using RNNs, CNNs, or Transformers.
- **Decoder:** Generates the output text sequence based on the encoder reprensentation. Current state-of-the-art models often use attention mechanisms to focus on relevant parts of the input during decoding.

ASR Benchmarking: Setup

- Accent and dialect variation
- Background noise handling
- Real-time processing requirements
- Domain-specific vocabulary

ASR Benchmarking: Results

- Accent and dialect variation
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ASR Benchmarking: Inferences

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MT: Overview

- Translates text from source to target language
- Evolution: Rule-based $\rightarrow$ Statistical $\rightarrow$ Neural
- Transformer architecture revolution
- Quality assessment and evaluation

MT: Key Approaches

- Neural Machine Translation (NMT)
- Transfer learning and multilingual models
- Low-resource language handling
- Context-aware translation

TTS: Overview

- Converts text into natural-sounding speech
- Components: Text analysis, prosody generation, synthesis
- Neural TTS models (WaveNet, Tacotron)
- Voice cloning and personalization

TTS: Quality Factors

- Naturalness and intelligibility
- Prosody and emotion
- Multi-speaker capabilities
- Real-time generation

End-to-End Pipeline

- ASR $\rightarrow$ MT $\rightarrow$ TTS workflow
- Use case: Real-time speech translation
- Latency and quality trade-offs
- Error propagation considerations

Pipeline Architecture

- Modular vs. end-to-end approaches
- API integration strategies
- Scalability and deployment
- Performance optimization

Applications

- International conference interpretation
- Language learning platforms
- Accessibility tools
- Customer service automation

Conclusion

- Integrated ASR-MT-TTS systems enable powerful applications
- Continued advances in neural architectures
- Challenges remain in low-resource scenarios
- Future directions: multimodal integration, personalization

Thank You

Thank you for your attention!

Questions?

Contact: your.email@example.com